

Dispatch System Design

WORKFLOWS, ARCHITECTURE, AND IMPLEMENTATION

TRACE — Dispatch & Patrol Design Document

The Patroller (Reporter in the field)

Driving around an assigned area. Phone mounted on the dash or in a cupholder.

One hand free, sometimes none. Daylight and nighttime. They are watching traffic, not their phone. When something catches their eye, they need to act in seconds, not minutes.

The Dispatch Operator

Sitting at a desk with a laptop. Managing multiple patrollers simultaneously.

Fielding phone calls from community members. Making quick decisions: is this vehicle in our system? Do I send someone? Who is closest? They are the air traffic controller.

The Community Caller (outside the app)

A resident who sees something and calls in. They describe a vehicle and a location. They do not use TRACE. The operator translates their call into a dispatch event.

What happens now (group chat)

Patroller texts: "Silver Honda Civic, plate ABC-1234, heading east on Oak St."

Dispatch checks the plate. Texts back: "That's in our system, confirmed."

Dispatch texts the group: "Confirmed vehicle at Oak and 3rd, who can respond?"

Another patroller texts: "On my way."

What TRACE should do

Step 1: Submit (3 seconds)

The Report screen opens with the plate field focused and the keyboard up.

GPS is captured automatically. The patroller types the plate, taps a direction arrow, and hits Submit. Photo is optional, one tap to capture.

They do not describe the location in words. The GPS does it.

If the patroller starts typing a plate already in the database, the app suggests it: "Match: ABC-1234 (Silver Honda Civic)." One tap to select.

Fewer typos, faster submission, automatic vehicle linking.

Step 2: Receipt with live status

After submit, the screen does not just say "Submitted." It shows a status card that updates in real time:

```
✓ Submitted                10:42:03 AM
■ Awaiting dispatch review
```

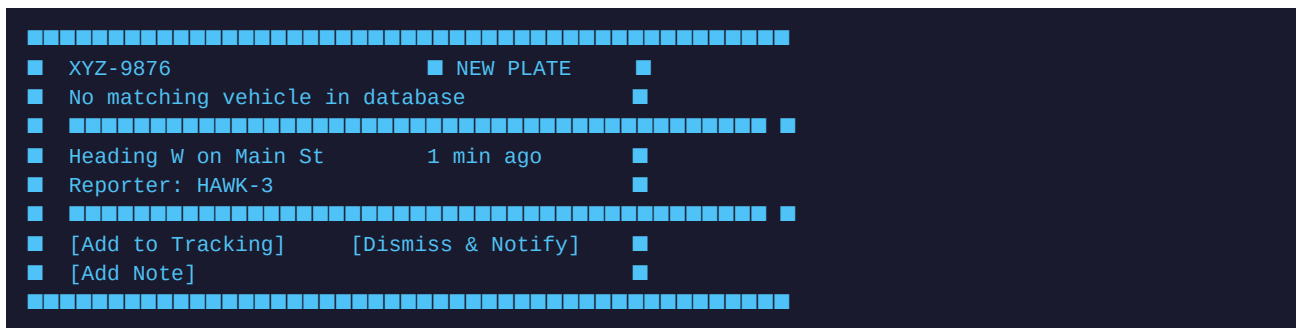
Then, as the operator acts:

```
✓ Submitted                10:42:03 AM
✓ Plate checked            10:42:15 AM
✓ Confirmed – patrollers dispatched
```

Or:

```
✓ Submitted                10:42:03 AM
✓ Plate checked            10:42:15 AM
– Not in our database. No action needed.
```

This is the group chat workflow, structured. The patroller knows their report was received, processed, and what happened next. No ambiguity.



1. Marks the sighting as confirmed
2. Opens the dispatch panel (slide-in, not a new page)
3. Shows a map centered on the sighting location
4. Lists available reporters with their last-known distance (if sharing location)
5. Operator selects who to dispatch (one, several, or all)
6. Adds optional notes ("Silver Civic, two occupants, parked at gas station")
7. Taps "Send Dispatch"

The original reporter gets: "Confirmed. Patrollers dispatched."

The dispatched patrollers get: push notification with location and details.

The dispatch event appears on both operator and reporter maps.

"Dismiss & Notify"

Tapping this:

1. Marks the sighting as reviewed
2. Sends the original reporter: "Not in our database. No action needed."
3. The sighting is filed, not deleted. New plates that appear multiple times across different days will surface organically in future reviews.

"Add to Tracking"

For new plates the operator wants to watch:

1. Creates a vehicle entry with the plate, description, and initial suspicion level
2. Optionally dispatches (same flow as Confirm & Dispatch)
3. Future sightings of this plate will auto-match

The operator receives a phone call: "There's a white van idling on Elm St, been there for an hour, two men sitting inside."

There is no sighting in TRACE. The operator creates a dispatch from scratch.

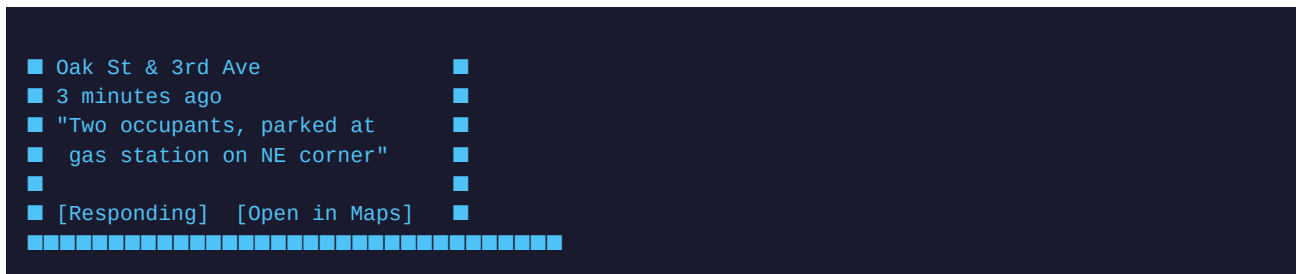
Dispatch from the Intel Map

1. Operator clicks a location on the map
2. A form appears:

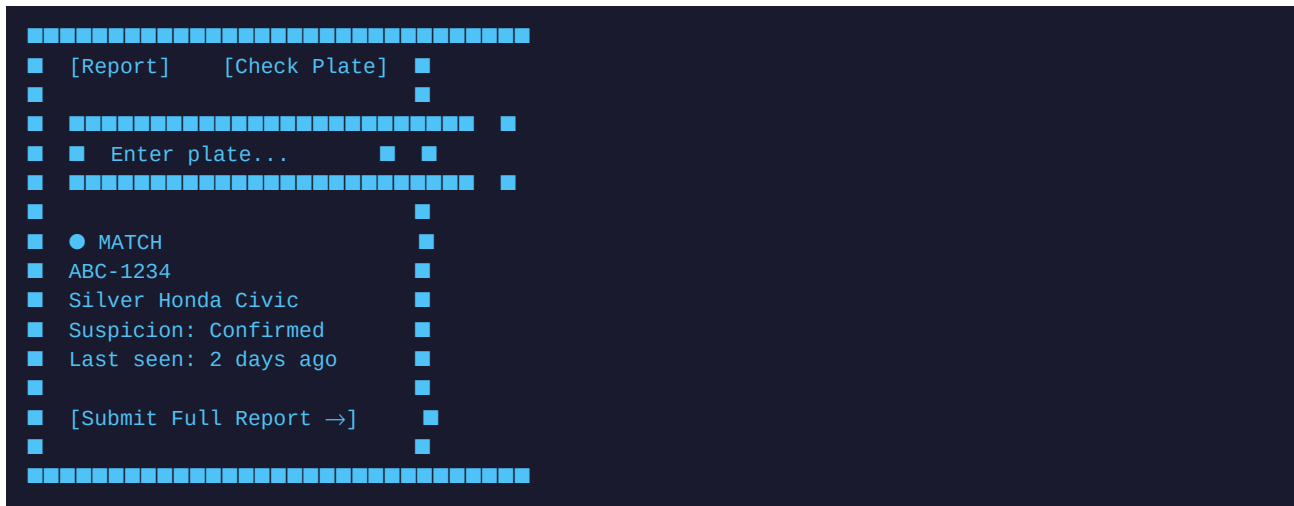
```
Event Type: [Suspicious Vehicle ■]
Priority:   [● Urgent ■ Routine ■ Info]
Notes:     White van, idling 1hr, two occupants
Source:    Community call
           ████████████████████████████████
           [Dispatch to: FALCON-7, HAWK-3 ■]
           [Send Dispatch]
```

TRACE - Dispatch
Confirmed vehicle at Oak St & 3rd Ave
Silver Honda Civic (ABC-1234) - Priority
Tap to view on map

[illegible]



A toggle or separate mode on the Report screen:



If the plate matches, the patroller can escalate to a full sighting report with one tap (pre-fills the plate and vehicle info). If no match, they move on.

This replaces texting dispatch "can you run this plate?"

What it shows

- Active dispatch pins (colored by priority, event type icons)
- The patroller's own sightings from the current shift (small, muted)
- The patroller's current location (blue dot)
- Satellite view by default (matches the real world they are looking at)

What it does NOT show

- Heatmaps, corridors, co-occurrence zones (operator-level intelligence)
- Other reporters' sightings (reporter-to-reporter invisibility)
- Vehicle suspicion levels or full dossiers
- Actor profiles

If a reporter's device is taken, the adversary sees dispatch pins (which are time-limited and auto-expire) and the reporter's own sightings. They do not see the operational intelligence picture.

Dispatch pin lifecycle on the map

- NEW: pin pulses briefly, colored border
- ACTIVE: solid pin, event type icon, priority color
- RESPONDING: pin shows who is responding ("FALCON-7 en route")
- CLOSED: pin fades and disappears after a short delay

Stale dispatch auto-close

Dispatches older than a configurable threshold (default: 4 hours) auto-close.

The map stays clean. Closed dispatches are in the operator's history.

1. Plate input auto-suggest

When the reporter types a plate, if it matches a vehicle in the database, show the match inline. "ABC-1 → ABC-1234 (Silver Honda Civic)." Tap to auto-fill. Reduces errors, speeds entry, auto-links to the right vehicle.

2. Submission vibration

A short, distinct vibration on successful submit. Not a sound (conspicuous).

The patroller feels the confirmation without looking at the screen.

3. Dispatch vibration pattern

Different from the submission vibration. Two quick pulses = dispatch alert.

The patroller learns the pattern and knows what it means without reading the screen.

4. Time-since on everything

Every timestamp shows relative time: "2 min ago", "1h ago." Absolute times are secondary. In field operations, "3 minutes ago" matters more than "10:42:03 AM."

5. Night mode auto-switch

Between sunset and sunrise (based on GPS location), auto-switch to dark mode.

A bright screen in a dark car is visible from outside and ruins night vision.

6. Report screen returns to ready state

After submitting, the status card shows for 30 seconds, then the screen resets to the plate input, focused and ready. The patroller should never have to navigate back to the report screen manually.

7. Closest-reporter indicator

When the operator is dispatching, show which reporters are nearest to the location. This is opt-in (reporters must consent to location sharing).

If enabled, the dispatch panel shows: "FALCON-7 (0.8 mi), HAWK-3 (2.1 mi)."

8. Dispatch outcome tracking

Every dispatch records: response time (how long until first responder acknowledged), arrival time (how long until on scene), outcome (sighting confirmed / nothing found / suspect fled). This is operational analytics the chapter can review weekly.

9. Shift concept

Patrollers work shifts. A "start shift / end shift" toggle that:

- Begins tracking their sightings for the current session
- Shows only current-shift sightings on the reporter map
- Enables location sharing (for proximity dispatch)
- Disables the dead man's switch timer during active shift

This keeps the map clean and the analytics per-shift.

10. Audio cue option

For patrollers who want it, a brief audio tone on dispatch. Different tones for different priorities. Configurable in settings, off by default.

11. Dispatch templates

Common dispatch types the operator sends repeatedly: "Confirmed vehicle, respond," "Community call, check area," "Plate swap suspected, investigate."

One-tap templates that pre-fill the dispatch form.

12. The "nothing happened" report

When a patroller checks out a dispatch and finds nothing, that is still intelligence. The "Nothing found" report should be easy to submit (one tap) and should record the time and location. Over time, patterns of false positives at certain locations are useful data.

The chapter admin defines event types. Defaults:

Type	Icon	Default Priority	Description
Confirmed Vehicle	car	Urgent	Known vehicle from the database
Community Report	radio	Routine	Called in by a community member
Area Check	compass	Routine	General area to patrol
Plate Swap Alert	alert-triangle	Urgent	Vehicle suspected of changing plates
Suspicious Activity	eye	Routine	Non-vehicle activity to investigate

Each type has:

- Label (editable)
- Icon (from curated set: ~15 options)
- Color (for map pin)
- Default priority (overridable per dispatch)
- Auto-close timer (hours, 0 = manual close only)

dispatch_event_types

Chapter-configurable. Same pattern as vehicle_types.

id, chapter_id, label, icon, color, default_priority, auto_close_hours, sort_order, created_at

dispatch_events

The core dispatch record.

id, chapter_id, sighting_id (nullable), event_type_id, lat, lng, location_description, notes, source (sighting|community_call|operator), priority (urgent|routine|info), status (open|responding|on_scene|closed|expired), created_by, created_at, closed_at, close_reason

dispatch_assignments

Who was dispatched.

id, dispatch_event_id, reporter_id, status (assigned|responding|on_scene|completed|declined), assigned_at, responded_at, arrived_at, completed_at, notes

dispatch_outcomes

What happened.

id, dispatch_event_id, reporter_id, sighting_id (nullable), outcome (confirmed|not_found|suspect_fled|false_alarm|other), notes, created_at

sighting_feedback

Feedback pushed to the original reporter.

id, sighting_id, message, feedback_type (confirmed|dismissed|info), sent_at, read_at

Phase 1: Auto plate lookup + reporter feedback

Highest value, least code. Enhance triage cards with automatic plate matching. Add "Confirm & Dispatch" and "Dismiss & Notify" actions that push feedback to the reporter. Reporter sees live status on submissions.

Phase 2: Dispatch events + reporter map

Build the dispatch table, operator dispatch panel (from triage and from map), reporter map tab with active pins. Push notifications for dispatch.

Phase 3: Quick plate check + dispatch responses

Plate check mode on reporter. "Responding" / "On scene" / "Nothing found" lifecycle. Dispatch outcome tracking. Operational analytics.

Phase 4: Polish

Plate auto-suggest, proximity dispatch, night mode auto-switch, shift concept, dispatch templates, vibration patterns, audio cues.